

REINFORCE

Monte-Carlo Policy Gradient and Variance Reduction

Reinforcement Learning

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1 Setup

We work in the episodic setting. The objective and its gradient are quoted from the policy gradient theorem reports:

$$J(\theta) = \mathbb{E}_{\tau \sim p_\theta} [G_0], \quad \nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} \left[\sum_{t=0}^{T-1} \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t | S_t) \right].$$

We solve $\theta^* = \arg \max_\theta J(\theta)$ by the stochastic-approximation recursion $\theta_{k+1} = \theta_k + \alpha_k \hat{g}_k$ and read each estimator through the decomposition $\hat{g}_k = g_k + b_k + \varepsilon_k$. This report develops two Monte-Carlo estimators, both unbiased ($b_k = 0$): plain REINFORCE (high variance), then the same estimator with a baseline (a control variate that cuts the variance for free).

Part I REINFORCE (Reward-to-Go)

2 The Estimator

2.1 From the Exact Gradient to a Sampled Estimator

The exact gradient is an expectation over trajectories,

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta} \left[\sum_{t=0}^{T-1} \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t | S_t) \right],$$

which we cannot evaluate: it averages over every trajectory the policy could generate. The one move is to replace the expectation by its sample analogue. Draw a trajectory by running the current policy,

$$\tau = (S_0, A_0, R_1, S_1, \dots, S_{T-1}, A_{T-1}, R_T, S_T) \sim p_\theta,$$

and drop the expectation, keeping the integrand evaluated on that single draw. This gives the single-sample (single-trajectory) estimator

Key Result

$$\hat{g} = \sum_{t=0}^{T-1} \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t | S_t), \quad G_t = \sum_{k=0}^{T-1-t} \gamma^k R_{t+k+1}.$$

One trajectory in, one gradient vector $\hat{g} \in \mathbb{R}^{d'}$ out.

For completeness, the same construction with N independent trajectories $\{\tau^{(i)}\}_{i=1}^N$, each run under the current policy, gives the multiple-trajectory estimator

$$\hat{g}_N = \frac{1}{N} \sum_{i=1}^N \sum_{t=0}^{T_i-1} \gamma^t G_t^{(i)} \nabla_\theta \log \pi_\theta(A_t^{(i)} | S_t^{(i)}),$$

the plain average of N single-trajectory estimators. It has the same expectation and $1/N$ times the variance; the single-sample case $N = 1$ is the one used in the algorithm, and all analysis below is stated for it.

2.2 Monte-Carlo Nature

The weight G_t is the *sampled return*: the actual discounted sum of rewards observed from step t to the end of the episode. Forming it requires the whole tail of the trajectory, so the estimator is available only once the episode terminates, and the update is therefore per-episode. This is what makes REINFORCE a Monte-Carlo method: the weight on the score is a sampled return, not a bootstrapped estimate.

3 Properties

3.1 Bias

The estimator is unbiased. Taking the expectation over the trajectory recovers the exact gradient by construction, since $\hat{g}$ is the integrand of $\nabla_\theta J(\theta)$ evaluated on one draw:

$$\mathbb{E}_{\tau \sim p_\theta}[\hat{g}] = \mathbb{E}_{\tau \sim p_\theta} \left[\sum_{t=0}^{T-1} \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t | S_t) \right] = \nabla_\theta J(\theta).$$

In the decomposition $\hat{g}_k = g_k + b_k + \varepsilon_k$ this is $b_k = 0$: the noise $\varepsilon_k = \hat{g} - \nabla_\theta J$ is zero-mean, and there is no bias term.

3.2 Variance

Since the estimator is unbiased, its variance is its entire error, and it is the quantity the rest of the report attacks. Write the estimator as a sum of per-step terms,

$$\hat{g} = \sum_{t=0}^{T-1} \gamma^t G_t \nabla_\theta \log \pi_\theta(A_t | S_t),$$

and look at where the randomness enters. Each term is a product of two random factors: the return G_t (a scalar) and the score $\nabla_\theta \log \pi_\theta(A_t | S_t)$ (a vector), both random because the trajectory is random. Two things make the variance large.

The return is a noisy weight. $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$ is a sum over the entire future of the episode, and that future is itself random: a different sampled continuation gives a different G_t . Its spread grows with the reward magnitudes and with how far the horizon extends, so a single G_t can sit far from its mean $V^{\pi_\theta}(S_t)$. This noisy scalar multiplies the score, so the whole term inherits the return’s spread.

The terms accumulate. $\hat{g}$ adds T such noisy terms. Even discounted by γ^t , a long episode contributes many fluctuating pieces, and their variances add up (they are correlated through the shared trajectory, but the total still grows with the number of terms). A longer horizon therefore means both larger individual returns and more of them.

Putting these together, the variance of $\hat{g}$ grows with the horizon, with the reward scale, and with how noisy the score is (roughly, the size of the action space). Concretely, the offending factor in each term is the raw return G_t : it is large and fluctuating even when the *choice* at step t was unremarkable, because G_t carries the entire downstream outcome, most of which had nothing to do with the action taken at t . Part II removes exactly this excess: it replaces G_t by how much better than average the outcome was, $G_t - V^{\pi_\theta}(S_t)$, which fluctuates much less while leaving the mean untouched.

3.3 Stochastic-Approximation Reading

In the recursion $\theta_{k+1} = \theta_k + \alpha_k \hat{g}_k$, the drift is the true gradient (unbiased), the fixed point is a stationary point $\nabla_{\theta} J(\theta) = 0$, and the noise ε_k is zero-mean with the high variance above. Convergence to a stationary point follows under the standing Robbins-Monro step-size conditions; the high variance is what governs the rate.

4 Algorithm

Algorithm 1: REINFORCE (Monte-Carlo policy gradient)

Input: differentiable policy π_{θ} , step size α , discount γ

Output: learned parameters θ

initialize θ

repeat

 generate an episode $\tau = (S_0, A_0, R_1, \dots, A_{T-1}, R_T, S_T) \sim \pi_{\theta}$

for $t = 0, 1, \dots, T-1$ **do**

$G_t \leftarrow \sum_{k=0}^{T-1-t} \gamma^k R_{t+k+1}$

$\hat{g} \leftarrow \sum_{t=0}^{T-1} \gamma^t G_t \nabla_{\theta} \log \pi_{\theta}(A_t | S_t)$

$\theta \leftarrow \theta + \alpha \hat{g}$

until *converged*

The update is per-episode: collect a full trajectory, compute every reward-to-go G_t , form the single estimator $\hat{g}$, and take one ascent step. This is unbiased and correct; its only weakness is the variance of $\hat{g}$, which Part II reduces.

Part II REINFORCE with a Baseline

5 What a Control Variate Is

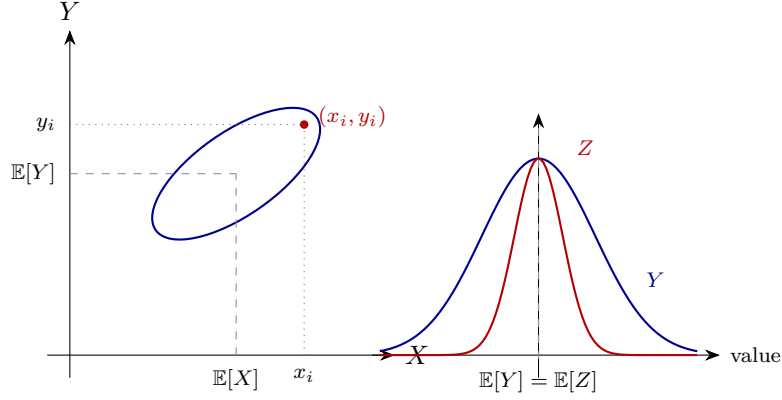
The baseline in Part II is a *control variate*. Full derivation in the Monte-Carlo Statistical Methods report.¹ The mechanism in one example.

We want the average *height* of a population and already know its average *weight*. On each sampled person we record both, $(x_i, y_i) = (\text{weight}, \text{height})$. Height and weight are correlated, so if a sampled person's weight lands above the known average weight, their height is probably above average too. We use that: subtract the weight-predicted part of the height deviation. The average is untouched (we subtract only a known-mean quantity), but the spread shrinks. Stronger correlation, more removed.

5.1 A Picture

Left: a one-sigma contour of $(X, Y) = (\text{weight}, \text{height})$, the tilt is the correlation. Right: the corrected variable Z sits at the same mean as Y but far narrower.

¹https://github.com/SaiSampathKedari/MonteCarlo-Statistical-Methods/blob/master/reports/ch04_01_ControlVariate_Foundations-and-Intuition.pdf



5.2 The Construction

Decompose Y into its best linear predictor from X plus an uncorrelated residual, $Y = aX + b + \epsilon$, $a = \text{Cov}(X, Y)/\text{Var}(X)$. Centering and subtracting the predictable part gives the corrected variable:

Key Result

$$Z = Y - a(X - \mathbb{E}[X]), \quad \mathbb{E}[Z] = \mathbb{E}[Y], \quad \text{Var}(Z) = (1 - \rho^2) \text{Var}(Y), \quad \rho = \text{Corr}(X, Y).$$

Same mean (the subtracted term has known mean zero), variance scaled by $(1 - \rho^2)$: at $\rho^2 = 1$ it vanishes, at $\rho = 0$ it is $Z = Y$, so it never hurts. The three requirements are a control variate X that is correlated with Y , has known mean $\mathbb{E}[X]$, and is sampled jointly with Y .

6 The Baseline as a Control Variate

Our estimator plays the role of Y , and we build a control variate X from a state-only function $b(S_t)$:

$$Y = \sum_{t=0}^{T-1} \gamma^t G_t \nabla_{\theta} \log \pi_{\theta}(A_t | S_t), \quad X = \sum_{t=0}^{T-1} \gamma^t b(S_t) \nabla_{\theta} \log \pi_{\theta}(A_t | S_t).$$

6.1 The Control Variate Has Known Mean Zero

The mean of X is exactly zero, so we do not even need to know it, it is pinned to zero for free:

$$\mathbb{E}[X] = \sum_{t=0}^{T-1} \gamma^t \mathbb{E}_{\tau} [b(S_t) \nabla_{\theta} \log \pi_{\theta}(A_t | S_t)] = \sum_{t=0}^{T-1} \gamma^t \mathbb{E}_{S_t} [b(S_t) \mathbb{E}_{A_t|S_t} [\nabla_{\theta} \log \pi_{\theta}(A_t | S_t)]] = 0,$$

because the inner expected score is zero, $\mathbb{E}_{A_t|S_t} [\nabla_{\theta} \log \pi_{\theta}(A_t | S_t)] = 0$. Two things make this work: the baseline $b(S_t)$ depends on the *state only*, so it pulls out of the inner expectation over the action; and the expected score vanishes.

Why the expected score is zero (mass conservation). *The policy is a probability distribution over actions, so its total mass is 1 for every θ . Moving θ by $d\theta$ adds probability to some actions and removes the same amount from others, but the total stays pinned at 1. The expected score asks, on average, how much mass a nudge $d\theta$ creates, and since the total can never change from 1, that average is zero.*

So this is a control variate whose mean is known exactly to be zero, which is the only condition it must satisfy, provided $b(S_t)$ never looks at the action A_t . If it did, it could not be pulled out of the action average and the mean would not vanish.

6.2 Which Baseline: the Value Function

Among all valid state-only baselines we take

$$b(S_t) = V^{\pi_\theta}(S_t),$$

the value function. Here is the geometric reason it is a good choice. Picture the states on a plane and V^{π_θ} as a surface over it. Fix a state s . Many different trajectories pass through s , and each collects a *different* random return G_t from s onward, so above the point s the returns form a vertical cloud of values, one per trajectory. By definition $V^{\pi_\theta}(s) = \mathbb{E}[G_t | s]$ is the *center* of that cloud: the surface threads through the middle of the scatter at every state.

Now the weight in the estimator changes from G_t to $G_t - V^{\pi_\theta}(S_t)$, which is each trajectory’s dot measured from the center of its cloud. Without the subtraction, if s is a high-value state the whole cloud floats high, so every trajectory through s carries a large weight of the same sign and pushes the same way, that large common offset is there because the *state* was good, not because the *action* was, and it is pure noise for judging the action. Subtracting the surface slides the cloud down to zero: trajectories that beat the local average get a positive weight, those that fell short get a negative one, and only this deviation, the genuine action-level signal, remains. The mean is untouched (the subtracted term has mean zero), but the weight has shrunk from “around $V^{\pi_\theta}(s)$ ” to “around zero,” and since the estimator’s variance scales with the size of the weight, the variance collapses. The centered weight $G_t - V^{\pi_\theta}(S_t)$ is the advantage: how much better this action did than the state’s average.

7 Estimating the Value Function

We do not know V^{π_θ} , so we estimate it. Fit a value predictor $\hat{V}_w(s)$ by Monte-Carlo regression on the returns already collected: each episode yields pairs (S_t, G_t) , and we minimize the squared error

$$\min_w \sum_t (G_t - \hat{V}_w(S_t))^2.$$

This is ordinary supervised regression of the observed return on the state, using the same rollouts as the policy update. The estimate is only *subtracted* as a baseline; the return G_t in the policy update stays the real sampled return, so the method remains Monte-Carlo. An imperfect $\hat{V}_w$ is safe: the mean-zero identity holds for *any* state function, so the gradient stays unbiased regardless of the fit; a poor fit merely reduces the variance less, and in the worst case ($\hat{V}_w$ uncorrelated with the return) recovers plain REINFORCE.

8 Algorithm

Algorithm 2: REINFORCE with baseline

Input: policy π_θ , value predictor $\hat{V}_w$, step sizes α, β , discount γ

Output: learned θ

initialize θ, w

repeat

 generate an episode $\tau = (S_0, A_0, R_1, \dots, A_{T-1}, R_T, S_T) \sim \pi_\theta$

for $t = 0, 1, \dots, T - 1$ **do**

$G_t \leftarrow \sum_{k=0}^{T-1-t} \gamma^k R_{t+k+1}$

$\delta_t \leftarrow G_t - \hat{V}_w(S_t)$ (advantage)

$w \leftarrow w + \beta \sum_{t=0}^{T-1} \delta_t \nabla_w \hat{V}_w(S_t)$ (fit the baseline)

$\theta \leftarrow \theta + \alpha \sum_{t=0}^{T-1} \gamma^t \delta_t \nabla_\theta \log \pi_\theta(A_t | S_t)$

until *converged*

Both updates use the real return G_t : the value predictor is fit to it by regression, and the policy is moved along the advantage-weighted score. The estimator stays unbiased and its variance is reduced. This closes the report; a fuller treatment of the control variate and the optimal baseline is left for a later revision.